

1.2 Prove by structural induction modulo renaming on ABTs that if $\mathcal{X} \subseteq \mathcal{Y}$, then, $\mathcal{B}[\mathcal{X}] \subseteq \mathcal{B}[\mathcal{Y}]$

Abstract Binding Tree definition:

1. If $x \in \mathcal{X}_s$, then $x \in \mathcal{B}[\mathcal{X}]_s$
2. For each operator o of arity $(\vec{s}_1.s_1, \dots, \vec{s}_n.s_n)s$, if for each $1 \leq i \leq n$ and each fresh renaming $\rho_i : \vec{x}_i \leftrightarrow \vec{x}'_i$, we have $\hat{\rho}_i(a_i) \in \mathcal{B}[\mathcal{X}, \vec{x}'_i]$, then $o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n) \in \mathcal{B}[\mathcal{X}]_s$.

Structural induction on $\mathcal{B}[\mathcal{X}]$: For property $\mathcal{P}[\mathcal{X}](a)$ if we show that both

1. If $x \in \mathcal{X}$, then $\mathcal{P}[\mathcal{X}](x)$
2. Let o be an operator with arity $(\vec{s}_1.s_1, \dots, \vec{s}_n.s_n)s$ and let $\vec{x}_1.a_1, \dots, \vec{x}_n.a_n$ be a sequence of abstractors with those valences. If for any $i \in [1, n]$ and any renaming $\rho_i : \vec{x}_i \leftrightarrow \vec{x}'_i$ with $\vec{x}'_i \notin \mathcal{X}$, $\mathcal{P}[\mathcal{X}](\hat{\rho}_i(a_i))$, then $\mathcal{P}[\mathcal{X}](o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n))$

Then by structural induction on $\mathcal{B}[\mathcal{X}]$, $\forall a \in \mathcal{B}[\mathcal{X}]$, $\mathcal{P}[\mathcal{X}](a)$

Let $\mathcal{P}[\mathcal{X}](a) = (\forall \mathcal{Y}' \supseteq \mathcal{X}, a \in \mathcal{B}[\mathcal{Y}'])$

The structural induction on $\mathcal{B}[\mathcal{X}]$ is given as follows:

1. Base case: $\forall x \in \mathcal{X} \forall \mathcal{Y}' \supseteq \mathcal{X}, x \in \mathcal{Y}'$ since $\mathcal{X} \subseteq \mathcal{Y}'$, so $x \in \mathcal{B}[\mathcal{Y}']$ by part 1 of the definition of $\mathcal{B}[\mathcal{Y}]$ and $\mathcal{P}[\mathcal{X}](x)$.
2. Inductive case: For given $\mathcal{Y}' \supseteq \mathcal{X}$, o of arity $(\vec{s}_1.s_1, \dots, \vec{s}_n.s_n)s$, and sequence of abstractors $\vec{x}_1.a_1, \dots, \vec{x}_n.a_n$ with corresponding valences, assume as the inductive hypothesis that for any $i \in [1, n]$ and any renaming $\rho_i : \vec{x}_i \leftrightarrow \vec{x}'_i$ with $\vec{x}'_i \notin \mathcal{X}$, that $\mathcal{P}[\mathcal{X}, \vec{x}'_i](\hat{\rho}_i(a_i))$. That is, $\forall \mathcal{Y}'' \supseteq (\mathcal{X}, \vec{x}'_i)$, $\hat{\rho}_i(a_i) \in \mathcal{B}[\mathcal{Y}'']$.

Fix arbitrary $\mathcal{Y}^{(3)} \supseteq \mathcal{X}$ so that any $\vec{x}'_i$ fresh for $\mathcal{Y}^{(3)}$ is also fresh for $\mathcal{X}$. Then, $\mathcal{Y}^{(3)}, \vec{x}'_i \supseteq \mathcal{X}, \vec{x}'_i$ is well-defined and true for any $\vec{x}'_i \notin \mathcal{Y}^{(3)}$.

Now we weaken our inductive hypothesis by quantifying over only $\vec{x}'_i \notin \mathcal{Y}^{(3)}$, giving $\forall i \in [1, n] \forall \hat{\rho}_i : \vec{x}_i \leftrightarrow \vec{x}'_i$ s.t. $\vec{x}'_i \notin \mathcal{Y}^{(3)}$, $\hat{\rho}_i(a_i) \in \mathcal{B}[\mathcal{Y}^{(3)}, \vec{x}'_i]$. Therefore $o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n) \in \mathcal{B}[\mathcal{Y}^{(3)}]$ by part 2 of the definition of $\mathcal{B}[\mathcal{Y}^{(3)}]$.

Quantifying over arbitrary $\mathcal{Y}^{(3)}$ now gives that $\forall \mathcal{Y}^{(3)} \supseteq \mathcal{X}$, $o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n) \in \mathcal{B}[\mathcal{Y}^{(3)}]$ and $\mathcal{P}[\mathcal{X}](o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n))$

Therefore, $\forall a \in \mathcal{B}[\mathcal{X}]$, $a \in \mathcal{B}[\mathcal{Y}]$ by structural induction on $\mathcal{B}[\mathcal{X}]$ and $\mathcal{B}[\mathcal{X}] \subseteq \mathcal{B}[\mathcal{Y}]$